

The Infinite-History Shift Obstruction

Spectral-Disk Pollution, Non-Schatten Powers, and Deceptive Finite Sections

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Abstract

The exact rectangular history maps of RH-173 suggest a square completion on an infinite memory space. Freezing the normalized first-block dynamics at $B = rA$ and writing $q = \sqrt{\eta} > 0$, the canonical completion is

$$\mathcal{L}(y_0, y_1, y_2, \dots) = (By_0, qy_0, qy_1, qy_2, \dots)$$

on $\ell^2(\mathbb{N}_0; \mathcal{Y})$. We determine its spectrum and trace-ideal status exactly.

If $\mathcal{Y}$ is finite dimensional, then

$$\sigma(\mathcal{L}) = \sigma(B) \cup \{z : |z| \leq q\}, \quad \sigma_e(\mathcal{L}) = \{z : |z| = q\}.$$

The tail is a weighted unilateral shift. Therefore $\mathcal{L}$ is not compact, belongs to no Schatten class, and no positive power $\mathcal{L}^m$ is compact or Schatten. The ordinary Fredholm determinants $\det(I - z\mathcal{L}^m)$ required by the moving-cloud route are consequently undefined.

Finite truncations conceal this obstruction: their shift tails are nilpotent and have spectrum only at zero. Nevertheless, for $0 < |z| < q$ their resolvent satisfies the exact vector lower bound

$$\|(zI - qS_L)^{-1}e_0\| = \frac{1}{|z|} \left(\sum_{k=0}^{L-1} (q/|z|)^{2k} \right)^{1/2},$$

which grows exponentially with L . A 15-case audit verifies this formula; at $L = 64$ the largest recorded lower bound exceeds $10^{39.9}$ although every finite shift eigenvalue is zero.

This is a rigorous no-go theorem for the direct canonical infinite-history shift as a Fredholm-determinant operator. It does not reject finite cyclic closures, weighted differences that cancel the shift, scattering objects, or direct transfer-space constructions. The natural next branch is finite cyclic memory.

1 Why square the rectangular cocycle?

RH-173 constructs exact maps $T_t : \mathcal{K}_t \rightarrow \mathcal{K}_{t+1}$, but a standard operator determinant requires a square operator on one fixed Hilbert space [4]. The most literal square completion stores the entire past in

$$\mathcal{H} = \ell^2(\mathbb{N}_0; \mathcal{Y}) = \mathcal{Y} \oplus \mathcal{Y} \oplus \dots$$

If the time-dependent normalization ratio is frozen at $r > 0$, the current block evolves by $B = rA$ and every older block shifts one position with weight $q = \sqrt{\eta}$. The resulting operator is

$$\mathcal{L}(y_0, y_1, y_2, \dots) = (By_0, qy_0, qy_1, qy_2, \dots). \quad (1)$$

The frozen model is already sufficient to decide the determinant question. If even one positive constant q forces a noncompact shift tail, changing B or fitting a finite packet cannot make the literal completion trace class.

2 Block form and invariant tail

Let S be the unilateral shift on $\ell^2(\mathbb{N}; \mathcal{Y})$,

$$S(y_1, y_2, \dots) = (0, y_1, y_2, \dots),$$

and let $E : \mathcal{Y} \rightarrow \ell^2(\mathbb{N}; \mathcal{Y})$ inject into the first tail coordinate. Relative to $\mathcal{H} = \mathcal{Y} \oplus \ell^2(\mathbb{N}; \mathcal{Y})$,

$$\mathcal{L} = \begin{pmatrix} B & 0 \\ qE & qS \end{pmatrix}. \quad (2)$$

The tail subspace $\{0\} \oplus \ell^2(\mathbb{N}; \mathcal{Y})$ is invariant and the restriction is qS . The coupling qE has rank $\dim \mathcal{Y}$.

3 Exact spectrum

Theorem 3.1 (Spectral disk of the history completion). *Suppose $\dim \mathcal{Y} < \infty$ and $q > 0$. Then*

$$\sigma(\mathcal{L}) = \sigma(B) \cup \overline{D(0, q)}. \quad (3)$$

Proof. If $z \notin \sigma(B) \cup \overline{D(0, q)}$, both $z - B$ and $z - qS$ are invertible. The triangular inverse formula applied to (2) shows that $z - \mathcal{L}$ is invertible. This proves one inclusion.

Let $|z| = q$. The unilateral shift has approximate eigenvectors at every point of its boundary circle. Embedding those vectors in the invariant tail gives approximate eigenvectors for $\mathcal{L}$, so $z \in \sigma(\mathcal{L})$.

Now let $|z| < q$. If $z - B$ is not surjective, neither is $z - \mathcal{L}$, because the first output coordinate is $(z - B)y_0$. If $z - B$ is invertible, use the adjoint. For any nonzero $v \in \mathcal{Y}$, the tail vector

$$h_v = (v, (\bar{z}/q)v, (\bar{z}/q)^2v, \dots)$$

belongs to ℓ^2 and satisfies $qS^*h_v = \bar{z}h_v$. Choose $x \in \mathcal{Y}$ solving

$$(B^* - \bar{z})x = -qv.$$

Then (x, h_v) is a nonzero eigenvector of $\mathcal{L}^*$ with eigenvalue $\bar{z}$. Hence $z \in \sigma(\mathcal{L})$. Finally, every point of $\sigma(B)$ makes the first-coordinate equation nonsurjective in finite dimension, so it also belongs to the spectrum. $\square$

The disk is independent of the detailed prime-dynamics block B . Its radius is fixed solely by the memory parameter. For the archived value $\eta = 1/512$,

$$q = \frac{1}{\sqrt{512}} = 0.0441941738 \dots$$

Small radius does not mean trace class: infinitely many directions carry the same nonzero shift weight.

4 Essential spectrum and Schatten obstruction

Theorem 4.1 (No compact or Schatten power). *For $q > 0$,*

$$\sigma_e(\mathcal{L}) = \{z : |z| = q\}. \quad (4)$$

Moreover, for every integer $m \geq 1$, $\mathcal{L}^m$ is not compact and does not belong to $\mathcal{S}_p$ for any $1 \leq p < \infty$.

Proof. The difference between $\mathcal{L}$ and $B \oplus qS$ is the finite-rank operator with only block qE . Essential spectrum is invariant under compact perturbations, the finite matrix B has empty essential spectrum, and the Fredholm essential spectrum of the unilateral shift is the unit circle [1]. This proves (4).

The invariant tail restriction of $\mathcal{L}^m$ is $q^m S^m$. The vectors $q^m S^m e_n$ form an orthogonal sequence of constant norm q^m , so the image of the unit ball is not relatively compact. Thus $\mathcal{L}^m$ is not compact. Every Schatten operator is compact, completing the proof [2]. $\square$

5 Which memory weights could evade the obstruction?

The failure is caused by a constant tail weight, not by the mere existence of history coordinates. This can be stated exactly for general weighted shifts. Let $(q_n)_{n \geq 0}$ be nonnegative and define

$$W(y_0, y_1, \dots) = (0, q_0 y_0, q_1 y_1, \dots)$$

on $\ell^2(\mathbb{N}_0; \mathcal{Y})$, where $d = \dim \mathcal{Y} < \infty$.

Theorem 5.1 (Compactness and Schatten criterion for age weights). *The weighted history shift W is compact if and only if $q_n \rightarrow 0$. For $1 \leq p < \infty$,*

$$W \in \mathcal{S}_p \iff \sum_{n \geq 0} q_n^p < \infty, \quad (5)$$

and in that case

$$\|W\|_p^p = d \sum_{n \geq 0} q_n^p. \quad (6)$$

Proof. The positive operator W^*W is block diagonal with blocks $q_n^2 I_{\mathcal{Y}}$. Hence the singular values of W are precisely the numbers q_n , each repeated d times. They tend to zero exactly when W is compact, and their p th powers are summable exactly under (5). $\square$

The archived memory recursion has the same shift factor $q_n \equiv \sqrt{\eta}$, so it lies at the opposite extreme: none of the singular values decays with age. A trace-class redesign would require, for example, age weights in ℓ^1 ; a Hilbert–Schmidt redesign would require weights in ℓ^2 . Such a redesign changes the original recursive memory and must be justified physically rather than inserted after the fact.

6 Explicit exterior resolvent

Outside the spectral union in Theorem 3.1, the triangular inverse is

$$(z - \mathcal{L})^{-1} = \begin{pmatrix} (z - B)^{-1} & 0 \\ (z - qS)^{-1}qE(z - B)^{-1} & (z - qS)^{-1} \end{pmatrix}. \quad (7)$$

Consequently, with $b(z) = \|(z - B)^{-1}\|$ and $s(z) = \|(z - qS)^{-1}\|$, a simple block-row estimate gives

$$\|(z - \mathcal{L})^{-1}\| \leq b(z) + s(z) + q b(z)s(z). \quad (8)$$

For $|z| > q$, $s(z) \leq (|z| - q)^{-1}$. Formula (7) makes clear that the first-block spectrum and the shift circle are the only exterior singularities; the finite-rank coupling creates no additional spectral component outside their union.

Corollary 6.1 (Direct Fredholm-determinant route is unavailable). *Neither $\det(I - z\mathcal{L})$ nor $\det(I - z\mathcal{L}^m)$ is an ordinary Fredholm determinant for any $m \geq 1$. In particular, squaring does not remove the history-tail obstruction.*

One could seek a relative determinant against the shift background, but that would be a scattering construction with a specified reference operator. It is not the direct moving-cloud determinant required in RH-80 [5].

7 Why finite spectra are misleading

Let S_L be the nilpotent unilateral shift on $\mathbb{C}^L$, $S_L e_j = e_{j+1}$ for $j < L - 1$ and $S_L e_{L-1} = 0$. Its spectrum is $\{0\}$ for every L , so eigenvalue plots cannot reveal the limiting disk.

Proposition 7.1 (Exact interior resolvent growth). *For $z \neq 0$,*

$$(zI - qS_L)^{-1}e_0 = \sum_{k=0}^{L-1} \frac{q^k}{z^{k+1}} e_k, \quad (9)$$

and hence

$$\|(zI - qS_L)^{-1}\| \geq \frac{1}{|z|} \left(\sum_{k=0}^{L-1} (q/|z|)^{2k} \right)^{1/2}. \quad (10)$$

For every fixed $0 < |z| < q$, the right side grows exponentially in L .

Proof. Because $S_L^L = 0$, the finite Neumann expansion is exact:

$$(zI - qS_L)^{-1} = z^{-1} \sum_{k=0}^{L-1} (qS_L/z)^k.$$

Applying it to e_0 gives (9). The basis vectors are orthonormal, so taking the norm gives the displayed lower bound. $\square$

This is a standard finite-section manifestation of nonnormal spectral pollution: the eigenvalues remain at zero while the pseudospectrum expands toward the shift disk [3].

8 Finite-section audit

The audit uses $L = 4, 8, 16, 32, 64$ and points $|z|/q = 0.25, 0.50, 0.75$, for 15 resolvent cases. Direct triangular solves agree with (10) to relative error below 10^{-12} . Every finite shift spectrum is exactly zero. At $L = 64$ the largest recorded resolvent-vector lower bound has

$$\log_{10} \|(zI - qS_L)^{-1}e_0\| = 39.9005.$$

The same audit forms square block truncations of (1) with $B = 0.8 \operatorname{diag}(0.25, -0.40)$. Exactly $2(L - 1)$ singular values remain at least q in every truncation, matching the growing shift multiplicity. This finite singular floor is the trace-ideal obstruction in elementary matrix form.

9 Scope of the no-go theorem

Theorem 4.1 rejects one precise branch:

literal infinite normalized history shift $\implies$ direct trace-ideal determinant.

It does not reject:

1. a finite cyclic closure, whose wrap edge replaces the unilateral tail;
2. a relative or scattering determinant with qS as reference;
3. a difference of two histories in which the shift cancels;
4. a compactly weighted memory with weights tending to zero by age;
5. a direct physical transfer-space Riesz construction.

The next paper develops the finite cyclic branch because it preserves a direct finite determinant and, unexpectedly, reproduces the geometric cloud polynomial exactly.

No physical cycle identification, all-level Riesz projection, complement limit, self-adjoint operator, zeta divisor identity, or Riemann Hypothesis is proved.

References

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